Philosopher's Stone:
A Deterministic Open-Text Instrument for Epistemic Octahedron Profiling

Architecture, Measurement Logic, Risk Memory, and Validation Program

Raul Miguel Pascual Paras III
Independent Conceptual Research

May 4, 2026

Abstract

This paper defines Philosopher's Stone as the first working open-text instrument built around the Epistemic Octahedron. The Epistemic Octahedron supplies the theoretical geometry: active worldview positions are represented on the octahedral surface $|x| + |y| + |z| = 1$, the coordinate origin is reserved for pre-philosophical nullity, the lower vertex represents epistemic collapse, and the upper vertex represents objective peak philosophical maturity. Philosopher's Stone operationalizes that theory by turning natural-language worldview fragments into structured evidence, deterministic semantic aggregation, gate-state memory, risk-state memory, and geometrically valid plotted points.

The system separates probabilistic interpretation from deterministic scoring. A language model is used only as an extractor under a strict contract. It returns structured JSON describing axis evidence, local principles, boundaries, tradeoffs, contradictions, gate events, and risk events. A deterministic profiler then normalizes this evidence, updates persistent state, computes semantic coordinates, applies scope and peak rules, handles structured risks, and projects active worldview positions onto the octahedral surface. This separation makes the instrument auditable: the language model does not directly choose the final coordinates.

The central claim of this paper is that Philosopher's Stone is no longer merely a visualizer or prompt experiment. It is a coherent candidate instrument for open-text epistemic profiling. Its current implementation includes gate-based epistemic stability tracking, structured risk events, low-signal null handling, determinate negative-evidence projection, absence-based negative guards, automatic gate-support validation, scope-relative peak detection, and peak-persistence logic. The system remains an instrument specification rather than a finished psychometric standard, but it now has enough structure to support benchmark construction, reliability testing, extractor comparison, and longitudinal validation.

1 Purpose

This document is the system paper for Philosopher's Stone. It should be read as a companion to the theory paper defining the Epistemic Octahedron.

The theory paper defines the graph: its geometry, axis meanings, surface constraint, null origin,

lower vertex, upper vertex, and the distinction between pre-philosophical absence, active negative epistemic instability, and mature integration.

This paper defines the instrument: how a repository can take open text, extract epistemic evidence, compute deterministic semantic values, preserve profile memory, and plot a worldview fragment onto the Epistemic Octahedron without letting the language model directly invent a coordinate.

The purpose is not to replace the theory paper. The purpose is to show how the theory becomes measurable in practice.

2 Core System Claim

The core system claim is:

> Philosopher's Stone is a deterministic open-text instrument for applying the Epistemic Octahedron to expressed worldview fragments. It separates language-model extraction from profiler computation, preserves the octahedral surface rule, distinguishes null from active worldview positions, tracks epistemic gates and risk states across compiles, and produces auditable coordinates rather than direct language-model judgment.

This is the proper status of the current repository. It is not merely a 3D graph viewer. It is not merely a personality quiz. It is not merely a prompt. It is an interpretation pipeline.

The system can score one thought, one stance, a worldview fragment, or eventually a fuller profile. This makes it useful both for casual philosophical testing and for more formal benchmark work.

3 Relation to the Epistemic Octahedron Theory

The Epistemic Octahedron theory fixes the following elements:

- active worldview positions lie on the octahedral surface,
- the surface rule is $|x| + |y| + |z| = 1$,
- the origin $(0, 0, 0)$ is the pre-philosophical null state,
- the lower vertex $(0, -1, 0)$ is epistemic collapse,
- the upper vertex $(0, 1, 0)$ is objective peak philosophical maturity,
- x represents the empathy/practicality tension,
- z represents the wisdom/knowledge tension,
- y represents negative versus positive epistemic stability.

The system does not redefine these meanings. It attempts to measure them.

This distinction is essential. The Epistemic Octahedron is the theoretical model. Philosopher's Stone is an instrument designed to apply the model to text. A theory can be coherent before a particular instrument is final. An instrument can improve while the theory remains stable.

4 System Overview

The current repository implements a layered pipeline:

1. The user enters a piece of worldview text.
2. `contracts.js` builds an LLM packet containing the input text, profiler memory, current canon, gate snapshot, and the extraction contract.
3. The LLM returns structured JSON. It does not compute final coordinates.
4. `profiler.js` normalizes the JSON, updates memory, computes semantic values, applies deterministic rules, and projects the point.
5. The visualizer renders the resulting point on the Epistemic Octahedron surface.

The strongest architectural feature is the separation between extraction and scoring. The LLM reads language. The profiler performs the measurement.

5 Dataflow

The system can be summarized as:

User text $\rightarrow$ LLM packet $\rightarrow$ structured JSON extraction $\rightarrow$ deterministic profiler $\rightarrow (a, s, b) \rightarrow (x, y, z) \rightarrow$ visualiz

The semantic triple is:

$$(a, s, b)$$

where a is the empathy/practicality semantic value, s is the epistemic-stability semantic value, and b is the wisdom/knowledge semantic value.

If the worldview is active enough, the profiler projects:

$$(x, y, z) = \frac{(a, s, b)}{|a| + |s| + |b|}.$$

If the semantic magnitude is not active enough for a fresh one-entry profile, the point can remain at the reserved null state:

$$(0, 0, 0).$$

This implements the theory's distinction between pre-surface nullity and active surface positions.

6 The Contract Layer

The contract layer is implemented mainly in `contracts.js`. Its job is to constrain the LLM.

The contract tells the model to extract structured evidence only. It forbids the model from computing final coordinates, maturity percentages, or projection math. This makes the LLM an evidence parser rather than the final authority.

The packet includes:

- the user's current text,
- current profile canon,
- suggested canon optimization,
- profile memory,
- structured risk memory,
- gate snapshot,
- the required JSON schema.

The required JSON output includes:

- analysis scope,
- scope strength,
- profile target frame,
- statement modes,
- semantic grid,
- dimension consideration,
- claim commitments,
- scope profile,
- local extraction,
- axis events,
- local positive and negative y signals,

- structured risk events,
- triggered gate events,
- gate update proposals,
- profile update signals,
- canon optimization,
- notes.

The contract's most important instruction is that silence is neutral. Missing caveats, missing examples, or missing discussion of context should not become negative evidence by themselves. A negative gate requires positive evidence of a reasoning failure.

7 Targeting

The system distinguishes the profiled target from the object being discussed.

For example, a user may criticize an institution, ideology, person, or group. The profiler should not automatically treat the criticized object's failure as a failure in the user's reasoning.

The target frame can be:

- authorial endorsement,
- self-description,
- described subject,
- cautionary example,
- quoted view,
- mixed or ambiguous.

This matters because epistemic stability is assigned to the profiled referent. If the author is endorsing a stance, then y concerns the author's stance. If the text describes someone else's collapse, the collapse belongs to the described subject, not automatically to the author.

8 The Profiler Layer

The profiler layer is implemented in `profiler.js`. It is the measurement engine.

The profiler performs the following tasks:

- validates and normalizes payloads,

- stores entry history,
- merges principles and boundaries into profile canon,
- tracks six gate states,
- tracks structured risk events,
- aggregates semantic grid values,
- computes lateral axis values,
- computes epistemic stability,
- applies scope-relative peak logic,
- applies low-signal and active-worldview rules,
- projects semantic values onto the octahedral surface,
- produces diagnostic math and supporting notes.

The profiler is deterministic. Given the same profile state and same JSON payload, it should produce the same result.

9 Semantic Aggregation

The profiler first aggregates dense semantic evidence into totals for:

- empathy,
- practicality,
- wisdom,
- knowledge,
- x integration,
- z integration,
- positive epistemic-stability evidence,
- negative epistemic-stability evidence.

For the x axis, empathy and practicality form opposing pole evidence. For the z axis, wisdom and knowledge form opposing pole evidence.

Integration evidence moderates lateral exaggeration. A text that strongly emphasizes practicality but also integrates empathy should not be treated the same as a text that rejects empathy. Likewise,

a wisdom-heavy text that preserves knowledge-contact should not be treated the same as vague intuition without factual grounding.

A compact representation of the lateral rule is:

$$a = \text{clamp}\left(\frac{(E-P)(1-\rho_x\lambda_I)}{\sigma_x}, -c, c\right),$$

$$b = \text{clamp}\left(\frac{(W-K)(1-\rho_z\lambda_I)}{\sigma_z}, -c, c\right),$$

where E is empathy mass, P is practicality mass, W is wisdom mass, K is knowledge mass, ρ_x and ρ_z are integration ratios, λ_I is integration influence, σ_x, σ_z are saturation constants, and c is the semantic overflow ceiling.

10 Epistemic Stability

Epistemic stability is the vertical dimension. It is the most important axis because the Epistemic Octahedron defines maturity primarily through positive epistemic stability and integrated lateral structure.

The profiler computes s from:

- positive local y evidence,
- negative local y evidence,
- integration bonus,
- unresolved lateral penalty,
- contradiction penalty,
- positive gate state,
- negative gate state,
- scope pressure.

A compact representation is:

$$s = \text{clamp}\left(\frac{Y^+ - Y^- + B_I - P_U - \alpha C}{\sigma_y} + \beta\overline{G^+} - \gamma\overline{G^-}, -c, c\right).$$

Here Y^+ is positive epistemic evidence, Y^- is negative epistemic evidence, B_I is integration bonus, P_U is unresolved lateral pressure, C is contradiction pressure, $\overline{G^+}$ is positive gate state, and $\overline{G^-}$ is negative gate state.

11 The Six Epistemic Gates

The profiler currently uses six gates:

Gate	Meaning
G1 Counter-consideration	Recognition of real opposing pressure, qualification, or category distinction.
G2 Non-strawman	Fair representation of an opposed or competing view.
G3 Self-correction	Ability to revise, acknowledge overstatement, state what would change the view, or avoid false certainty.
G4 Contradiction handling	Handling of tensions, contradictions, or tradeoffs.
G5 Reality contact	Contact with consequences, constraints, feasibility, incentives, causal mechanisms, outcomes, or truth/reality testing.
G6 Non-self-sealing	Avoidance of circular logic, immune certainty, and sealed conclusions.

The gates are not moral decorations. They are operational criteria for epistemic stability.

Each gate stores:

- score,
- status,
- positive event count,
- negative event count,
- last event time,
- last evidence span.

The current statuses include dormant, lean positive, established positive, strong positive, lean negative, established negative, and strong negative.

12 Gate Coverage

The profiler distinguishes epistemic stability from gate coverage.

A profile can have a positive s value while still having low coverage. This means it has positive evidence, but not enough breadth across the gate ontology to confidently treat it as mature across a wider scope.

The profiler computes:

$$y_{\text{coverage}} = \frac{\sum w_g^{\text{covered}}}{\sum w_g^{\text{all}}}.$$

This is one of the system's most important design choices. It prevents a thin but positive text from being confused with a broadly tested worldview.

13 Structured Risk Events

The current system tracks structured risk events in addition to plain risk notes.

A structured risk event can include:

```
{
  "risk": "false_certainty",
  "status": "active",
  "target": "self",
  "confidence_score_0_to_1": 0.85,
  "addresses_risk_id": "",
  "evidence_span_text": "I am perfect in every way",
  "repair_requirements": [
    "self_correction",
    "qualified_framing",
    "reality_contact"
  ]
}
```

The profiler stores this as a case file rather than as a disposable warning. The historical evidence is preserved even if the risk is later softened or cleared.

Possible statuses are:

- active,
- softened,
- cleared.

This allows diachronic repair. A later text can clear a prior false-certainty risk without needing to quote the earlier wording exactly, as long as it directly repairs the relevant failure pattern.

14 Risk Repair

Risk repair should not erase history. It changes active status.

A repaired risk should preserve:

- original risk type,
- original evidence span,
- entry that introduced the risk,

- repair requirements.

It should add:

- clearing entry,
- clearance confidence,
- clearance evidence span,
- clearance note.

This supports a mature interpretation of growth. The system does not pretend the earlier risk never happened. It records that the profile has dealt with it.

15 Low-Signal Projection and the Active Worldview Threshold

The Epistemic Octahedron paper reserves the origin for pre-philosophical nullity. The system operationalizes this through a low-signal projection guard.

A fresh one-entry profile with very low semantic magnitude can remain at:

$$(0, 0, 0).$$

This prevents weak or vague text from being artificially stretched onto the surface.

However, determinate negative evidence bypasses this guard. If a text clearly shows active false certainty, self-sealing logic, contradiction evasion, or another serious epistemic failure, it should not disappear into the null state. It should project into active negative stability.

This preserves the theory's core distinction:

$$\text{null absence} \neq \text{active collapse}.$$

16 Absence-Based Negative Guard

One of the most important recent improvements is the absence-based negative guard.

The LLM extractor sometimes confuses missing caveats with negative epistemic evidence. For example, a short moral stance may not discuss every exception, consequence, or cultural context. That absence may be a scope gap, but it is not automatically false certainty, reality detachment, self-sealing logic, or a failed gate.

The profiler now rejects or downgrades negative artifacts when their evidence is only absence-based, such as:

- no discussion of consequences,

- no engagement with feasibility,
- strong assertion without openness indicators,
- normative claim without reference to outcomes,
- broad claim without qualifiers.

Those may remain unresolved scope gaps. They should not become negative y pressure unless the text positively shows closure, contradiction, reality detachment, or refusal of correction.

This is necessary for consistency with the theory. Silence is not collapse. Missing detail is not delusion.

17 Automatic Gate-Support Validation

The LLM extractor may correctly fill out supporting JSON fields but forget to emit the corresponding gate event. The profiler therefore includes a narrow deterministic validator.

For G4, the profiler can auto-support contradiction handling when:

- `local_extraction.tradeoffs` is nonempty,
- and axis integration includes a fair tradeoff, integrated tension, or explicit balance,
- and the evidence has adequate strength and confidence,
- and there is no self-targeted negative pressure contradicting the support.

For G5, the profiler can auto-support reality contact when the extractor's own fields show concrete material reality-contact or epistemic reality-contact, such as answerability to truth, false-belief correction, or reality testing.

This is not inventing evidence. It is enforcing implications already present in the extractor's own JSON.

18 Scope-Relative Peak

The theory defines the upper vertex as objective peak philosophical maturity. The instrument must apply this carefully because the system can score one thought, one stance, a worldview fragment, or a fuller profile.

The profiler therefore supports scope-relative peak.

A text may reach peak within its submitted referent if:

- the scope is complete for the text,
- relevant dimensions are covered,

- relevant gates are satisfied,
- core gates and stabilizer gates are present,
- integration is strong enough,
- there is no self-targeted negative pressure,
- no relevant dimension is rejected or explicitly deprioritized.

This does not mean the whole person is certified as permanently mature. It means the submitted stance or worldview fragment reaches mature integration within the scope it opened.

This is consistent with the theory because the theory defines the geometry, while the instrument defines the referent being plotted.

19 Peak Persistence

Another important recent improvement is peak persistence.

Once a profile has reached exact peak, a later positive but laterally asymmetric compile should not pull it down merely because it emphasizes practicality, empathy, wisdom, or knowledge in a contextual way.

Peak can be lost through genuine peak-breaking evidence, such as:

- self-targeted negative evidence,
- negative gate event,
- active profiled risk,
- introduced contradiction,
- failed gate,
- explicit dimension rejection,
- unresolved widened scope.

Peak should not be lost because of harmless lateral emphasis alone.

This implements an important theoretical point: mature philosophy is not frozen neutrality. It can emphasize one dimension contextually without losing integration.

20 Projection

The final projection is:

$$(x, y, z) = \frac{(a, s, b)}{|a| + |s| + |b|}.$$

This produces a surface point satisfying:

$$|x| + |y| + |z| = 1.$$

The only exception is the reserved null state:

$$(x, y, z) = (0, 0, 0),$$

which is not a surface point and is used only when the worldview is not active enough to project, or when the semantic vector is truly empty.

The profiler therefore preserves both essential states:

- the surface for active worldview positions,
- the origin for pre-surface nullity.

21 Diagnostics and Auditability

The system produces diagnostics instead of only a final point. This is crucial.

A useful compile can expose:

- semantic parameters,
- semantic percentages,
- projected point,
- projected percentages,
- gate coverage,
- gate snapshot,
- structured risks,
- low-signal projection status,
- peak persistence status,
- auto-supported gate events,
- scope diagnostics,
- dimension consideration,

- supporting entry profiles.

This audit trail makes the system testable. If a result looks wrong, the user can ask whether the failure came from the LLM extractor, the contract, the deterministic profiler, the projection guard, gate coverage, scope logic, or peak rules.

22 Why This Is an Instrument

A mere visualizer lets a user place points on a graph.

A mere prompt asks a model to describe a person.

A questionnaire forces the user into predefined answers.

Philosopher's Stone does something different. It takes open text, extracts structured evidence, applies deterministic measurement logic, preserves profile state, and returns a graph-valid worldview coordinate.

That qualifies it as an instrument in the practical sense:

- it has a theoretical object,
- it has an input mode,
- it has a structured extraction layer,
- it has deterministic scoring,
- it has state memory,
- it has diagnostics,
- it has a defined output geometry,
- it can be benchmarked.

It is best described as a prototype open-text epistemic profiling instrument.

23 What Is Fixed and What Can Evolve

The following are fixed or near-fixed theory-level commitments:

- the octahedral surface,
- the null origin,
- the lower vertex as collapse,
- the upper vertex as peak,

- the empathy/practicality axis,
- the wisdom/knowledge axis,
- the negative/positive epistemic-stability axis,
- the distinction between nullity, instability, and mature integration.

The following are instrument-level components that can evolve:

- exact gate weights,
- exact signal weights,
- contract wording,
- LLM extractor selection,
- risk taxonomy,
- scope thresholds,
- peak thresholds,
- confidence thresholds,
- UI labels,
- benchmark methodology.

This is not a weakness. It is the correct relation between theory and instrument.

24 Known Failure Modes

The major known failure modes are:

1. **Extractor variance:** the same LLM may output slightly different JSON for the same text across runs.
2. **Cross-field inconsistency:** the extractor may list tradeoffs but forget to emit G4.
3. **Absence-based punishment:** the extractor may treat missing caveats as negative evidence.
4. **Scope confusion:** the extractor may treat a narrow stance as if it opened a much broader worldview.
5. **Target confusion:** the extractor may assign criticized-object failure to the authorial stance.
6. **Projection exaggeration:** low-magnitude semantic vectors can produce visually strong projections if not guarded.
7. **Overgenerous peak:** a model may over-recognize maturity in polished but thin text.

The current system has guardrails for many of these, but benchmark testing should target them directly.

25 Validation Program

The next stage is validation.

The minimum validation program should include:

- repeated-run tests on the same input,
- known-good peak examples,
- known-bad collapse examples,
- false-certainty examples,
- repaired-risk examples,
- scope-expansion examples,
- target-confusion examples,
- extractor comparison across LLMs,
- human-rater comparison,
- ablation tests for profiler rules,
- longitudinal tests across multiple compiles.

The key question is not whether every single output feels flattering or harsh. The key question is whether the system behaves consistently under known cases.

26 Recommended Benchmark Categories

A benchmark suite should include at least the following categories:

Category	**Purpose**
Null text	Should remain at origin or pre-surface.
Thin positive stance	Should produce low or moderate positive stability, not peak.
Mature scoped stance	Should reach high stability or scope-relative peak.
False certainty	Should project negative or carry active risk.
Self-sealing logic	Should lower G6 and create active risk.
Reality detachment	Should lower G5 only when detachment is positively evidenced.
Absence trap	Should not fail gates merely for missing caveats.
Tradeoff integration	Should trigger or auto-support G4.
Epistemic reality contact	Should support G5 when truth-testing is explicit.
Risk repair	Should soften or clear prior active risk while preserving history.

Category	Purpose
Peak persistence	Should preserve peak unless real breaking evidence appears.
Unresolved scope expansion	Should allow loss of peak when new scope is opened and not handled.
Targeting test	Should not assign criticized-object failure to the author.

27 Repository Document Stack

The repo should eventually contain four major documents:

1. **Theory paper:** *The Epistemic Octahedron.* This defines the discovery.
2. **System paper:** *Philosopher's Stone: A Deterministic Open-Text Instrument for Epistemic Octahedron Profiling.* This is the present document.
3. **Instrument specification:** a technical reference for `contracts.js`, `profiler.js`, JSON fields, gates, risk events, formulas, and thresholds.
4. **Benchmark and validation protocol:** a testing document with gold examples, expected outputs, tolerance bands, and reliability criteria.

This document should function as the intellectual backbone of the repo's system layer.

28 Scientific Significance

The scientific significance of Philosopher's Stone is that it attempts to turn philosophical maturity into an auditable measurement pipeline without collapsing it into a simple personality type, sentiment score, or ideological label.

The system is significant because it combines:

- a formal geometric theory,
- a natural-language input mode,
- a typed evidence-extraction contract,
- deterministic scoring,
- persistent epistemic memory,
- lower-half risk differentiation,
- surface-valid projection,
- explicit diagnostics.

This makes it unusually testable for a worldview-modeling system.

29 Conclusion

Philosopher's Stone operationalizes the Epistemic Octahedron as a working open-text profiling instrument. It takes expressed worldview text, extracts structured semantic evidence, applies deterministic profiling logic, tracks gates and risks across compiles, and projects active worldview positions onto a geometrically valid octahedral surface.

The system now preserves the theory's central distinction between null origin, active negative instability, and mature integration. It treats silence as neutral, distinguishes missing detail from collapse, preserves risk history, supports risk repair, and prevents harmless contextual emphasis from breaking peak maturity.

The result is a serious candidate instrument. It is not the final possible implementation, but it is strong enough to deserve system-level documentation, benchmark construction, and validation work.

If the Epistemic Octahedron is the discovery, then Philosopher's Stone is the first instrument built to make that discovery operational.

A Compact Formula Reference

$$|x| + |y| + |z| = 1$$

$$(x, y, z) = \frac{(a, s, b)}{|a| + |s| + |b|}$$

$$a = \text{clamp}\left(\frac{(E - P)(1 - \rho_x \lambda_I)}{\sigma_x}, -c, c\right)$$

$$b = \text{clamp}\left(\frac{(W - K)(1 - \rho_z \lambda_I)}{\sigma_z}, -c, c\right)$$

$$s = \text{clamp}\left(\frac{Y^+ - Y^- + B_I - P_U - \alpha C}{\sigma_y} + \beta\overline{G^+} - \gamma\overline{G^-}, -c, c\right)$$

$$y_{\text{coverage}} = \frac{\sum w_g^{\text{covered}}}{\sum w_g^{\text{all}}}$$

B Compact Instrument Claim

Philosopher's Stone is a deterministic open-text instrument for applying the Epistemic Octahedron to expressed worldview fragments. It separates LLM extraction from profiler scoring, preserves the theory's surface geometry, tracks epistemic gates and structured risks over time, and returns auditable coordinates rather than opaque model judgment.